

Regression Models (Theory)

Welcome to Regression Models! In this session, we dive into **Regression**, a core supervised learning technique used for predicting continuous numerical values (e.g., house prices, student scores, or sales figures). You will learn the underlying mechanics of Linear Regression, its mathematical and matrix representation, and how to evaluate model performance using MAE, MSE, and R^2 Score.

1. Mechanics of Linear Regression

Linear regression assumes a linear relationship between the input features (X) and the target output (y). The goal of the algorithm is to find the "best-fit" straight line (or hyperplane in multi-variable spaces) that minimizes the overall prediction error.

A. The Mathematical Equation

For a single feature x , the linear regression equation is expressed as:

$$y = \beta_0 + \beta_1 x + \epsilon$$

Where:

- y is the predicted target value.
- β_0 is the **intercept** (bias).
- β_1 is the **coefficient** (slope) for the feature.
- ϵ is the error term (difference between actual and predicted values).

For multiple features ($x_1, x_2, \dots, x_n$), the equation expands to:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

B. Matrix Form & Interpretation

In data science libraries like Scikit-Learn, models process data efficiently using matrix algebra. The matrix representation of multiple linear regression is:

$$Y = X\beta + \epsilon$$

- Y : A vector of target values of shape $(m \times 1)$.

- X : The design matrix of input features of shape $(m \times n + 1)$ (including a column of ones for the intercept).
- β : The vector of coefficients (parameters) to be learned, of shape $(n + 1 \times 1)$.

Understanding Intercept and Coefficients:

- **Intercept (β_0):** The expected value of y when all input features (x) are equal to zero.
 - **Coefficient ($\beta_1, \beta_2, \dots$):** Represents the expected change in the target y for a one-unit increase in that specific feature, holding all other features constant. For example, in a house price model, a coefficient of 50 for `sqft` means every extra square foot increases the estimated price by \$50.
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2. Model Evaluation Metrics

Once a regression model is trained, we need quantitative metrics to measure how accurately it predicts continuous outcomes on unseen test data.

A. Mean Absolute Error (MAE)

MAE measures the average magnitude of absolute errors between predictions ($\hat{y}$) and actual values (y). It treats all errors equally.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- *Interpretation:* Expressed in the exact same units as the target variable (e.g., dollars for house prices). Lower is better.

B. Mean Squared Error (MSE)

MSE measures the average of the squared differences between estimated and actual values. Because errors are squared, MSE heavily penalizes large outliers.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- *Interpretation:* Useful for ensuring the model doesn't make massive blunder predictions, though harder to interpret directly because units are squared.

C. R^2 Score (Coefficient of Determination)

The R^2 score represents the proportion of variance in the dependent variable that is predictable from the independent variables.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

(where $\bar{y}$ is the mean of the actual target values)

- *Interpretation:*
 - $R^2 = 1.0$: Perfect predictions.
 - $R^2 = 0.0$: The model performs no better than simply guessing the average target value ($\bar{y}$).
 - Negative values indicate an arbitrarily worse model.
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3. Reference Links & Further Reading

- **Scikit-Learn Linear Models Documentation:** Official guide on Linear Regression implementation in Python.
 - **Scikit-Learn Regression Metrics Guide:** Comprehensive overview of MAE, MSE, and R^2 scoring functions.
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